

# **Electrospun hollow fibres for muscle tissue engineering in a soft bioreactor chamber**

DPhil Transfer Report



**Marie Beatrix Kruth**

St. Peter's College

University of Oxford

Supervised by

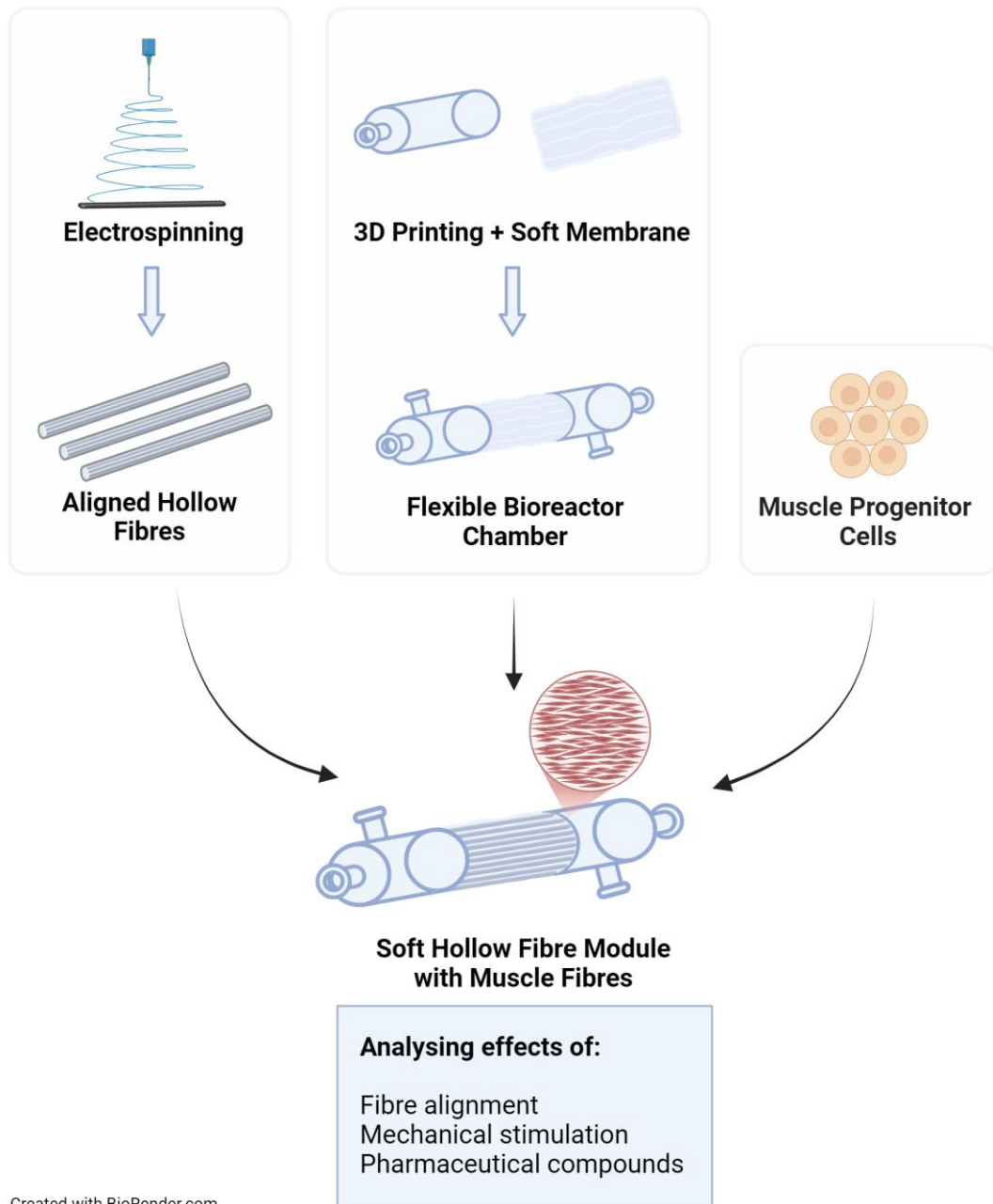
Prof. Pierre-Alexis Mouthuy

Prof. Philippa Hulley

Prof. Hua (Cathy) Ye

Submitted: Michaelmas 2024

This document is submitted to the Nuffield Department of Orthopaedics, Rheumatology, and Musculoskeletal Sciences (NDORMS) for the Transfer of Status requirement.



## Abbreviations

|                        |                                                         |
|------------------------|---------------------------------------------------------|
| <b>dECM</b>            | Decellularized Extracellular Matrix                     |
| <b>DMSO</b>            | Dimethyl Sulfoxide                                      |
| <b>ECM</b>             | Extracellular Matrix                                    |
| <b>eHFs</b>            | Electrospun Hollow Fibres                               |
| <b>F<sub>max</sub></b> | Maximum Force                                           |
| <b>HFs</b>             | Hollow Fibres                                           |
| <b>HFIP</b>            | Hexafluoroisopropanol                                   |
| <b>MHC</b>             | Myosin Heavy Chain                                      |
| <b>micro-CT</b>        | Micro-Computed Tomography                               |
| <b>MSK</b>             | Musculoskeletal                                         |
| <b>M<sub>w</sub></b>   | Molecular Weight                                        |
| <b>myoD</b>            | Myoblast Determination Protein 1,                       |
| <b>PCL</b>             | Polycaprolactone                                        |
| <b>PDO</b>             | Polydioxanone                                           |
| <b>PEDOT:PSS</b>       | Poly(3,4-ethylenedioxythiophene):Poly(styrenesulfonate) |
| <b>PVA</b>             | Polyvinyl Alcohol                                       |
| <b>qPCR</b>            | Quantitative Polymerase Chain Reaction                  |
| <b>RT</b>              | Room Temperature                                        |

## **List of Figures**

|                 |                                                                                           |             |
|-----------------|-------------------------------------------------------------------------------------------|-------------|
| <b>Figure 1</b> | Structure of skeletal muscle                                                              | <b>p.1</b>  |
| <b>Figure 2</b> | Hollow fibre bioreactor platforms                                                         | <b>p.3</b>  |
| <b>Figure 3</b> | Production of the dissolvable fibre collector                                             | <b>p.5</b>  |
| <b>Figure 4</b> | Production of electrospun hollow fibres                                                   | <b>p.6</b>  |
| <b>Figure 5</b> | Tensile mechanical properties of the dissolvable fibre collector                          | <b>p.7</b>  |
| <b>Figure 6</b> | Effects of PEDOT:PSS and PVA concentrations on electrical resistance and extrusion length | <b>p.8</b>  |
| <b>Figure 7</b> | Characterisation of pre- and post-heating fibre collectors made from different PVA types  | <b>p.9</b>  |
| <b>Figure 8</b> | Dimensions of pre- and post-stretch electrospun hollow fibres                             | <b>p.11</b> |
| <b>Figure 9</b> | Diameter and alignment of pre- and post-stretch electrospun microfibres                   | <b>p.12</b> |

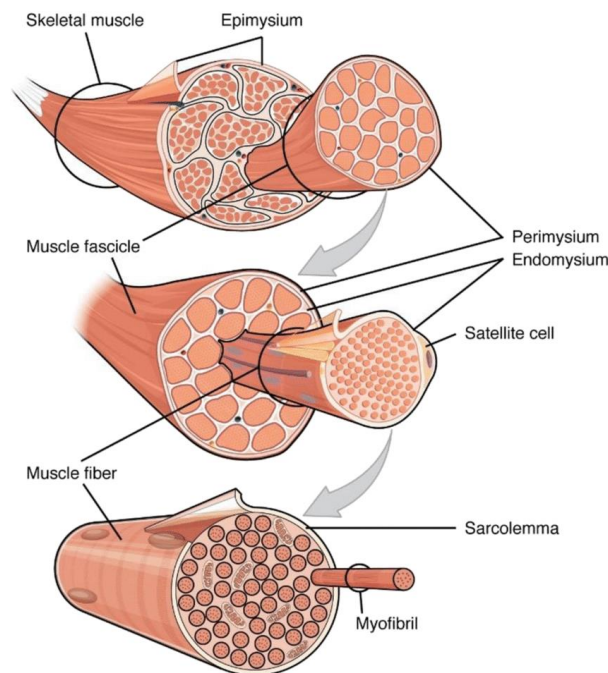
## Table of Contents

|                                                                                    |    |
|------------------------------------------------------------------------------------|----|
| 1. Background.....                                                                 | 1  |
| 1.1 Skeletal Muscle Research .....                                                 | 1  |
| 1.2 Skeletal muscle tissue engineering.....                                        | 2  |
| 1.3 Hollow fibre bioreactor platforms .....                                        | 2  |
| 1.4 Research objectives.....                                                       | 3  |
| 2. Materials and Methods .....                                                     | 4  |
| 2.1 Fabrication of a dissolvable fibre collector .....                             | 4  |
| 2.2 Characterisation of the dissolvable fibre collector .....                      | 5  |
| 2.3 Fabrication of electrospun hollow fibres .....                                 | 5  |
| 2.4 Characterisation of electrospun hollow fibres .....                            | 6  |
| 2.5 Image analysis .....                                                           | 6  |
| 3. Results and Discussion .....                                                    | 7  |
| 3.1 Production of a dissolvable fibre collector .....                              | 7  |
| 3.1.1 Effects of PVA concentration on wet-spinning and mechanical properties ..... | 7  |
| 3.1.2 Achieving electrical conductivity and large extrusion length .....           | 8  |
| 3.1.3 Effects of PVA type and heating on fibre collector properties .....          | 8  |
| 3.2 Production of electrospun hollow fibres.....                                   | 10 |
| 3.2.1 Suitability of the dissolvable fibre collector for electrospinning .....     | 10 |
| 3.2.2 Effects of draw speed on electrospun hollow fibre wall thickness.....        | 10 |
| 3.2.3 Effects of stretching on electrospun hollow fibre dimensions .....           | 10 |
| 3.2.4 Effects of stretching on microfibre diameter and alignment.....              | 11 |
| 3.2.5 Removal of the dissolvable collector core .....                              | 12 |
| 3.3 Limitations .....                                                              | 12 |
| 4. Conclusions and Future Work.....                                                | 13 |
| 4.1 Conclusions .....                                                              | 13 |
| 4.2 Future Work.....                                                               | 13 |
| 4.2.1 Electrospun hollow fibre scaffold .....                                      | 13 |
| 4.2.2 The bioreactor chamber and mechanical loading.....                           | 13 |
| 4.2.3 Functionality and utility of the muscle tissue construct.....                | 14 |
| 4.2.4 Gantt Chart .....                                                            | 15 |
| 4.2.5 Risk mitigation plan .....                                                   | 16 |
| 5. References.....                                                                 | 17 |
| 6. Appendix.....                                                                   | 20 |

# 1. Background

## 1.1 Skeletal Muscle Research

Skeletal muscle makes up 40% of human body mass and serves as the engine for movement and force generated through its contractile functions [1]. This tissue has a hierarchically structured architecture made up of bundles of muscle fascicles which consist of muscle fibres that are in turn composed of bundles of myofibrils [2] (Fig. 1). Muscle fibres can reach diameters of up to 100µm and lengths of up to 30 cm [3], and the tissue possesses remarkable regenerative capacities due to its resident satellite cells [4].



**Fig. 1: Structure of skeletal muscle**

Hierarchical structure showing bundles of muscle fascicles, muscle fibre and myofibrils [2].

Worldwide, musculoskeletal (MSK) conditions contribute to a significant global burden, with around 1.71 billion cases [5]. A subset of these conditions includes a wide spectrum of skeletal muscle diseases which contribute to pain and impairment of mobility, strength and function. This includes acquired disease forms, e.g. sarcopenia, as well as congenital forms, e.g. muscular dystrophy [6], [7]. Animal models have been extensively used in the study of the MSK system, including skeletal muscle. In the EU, over 70,000 animals are used annually for basic research on the MSK system in addition to over 48,000 animals used for translational and applied research on MSK disorders [8].

The use of animal models in MSK research is associated with major limitations, including reproducibility issues, ethical concerns, and limited relevance to human physiology. Reproducibility issues result from the complexity of animals as physiological systems, which can interfere with the replicability of experimental results [9]. Ethical concerns arise from the fact that animals may be subjected to different procedures which induce physical harm and cause severe pain, discomfort, and distress [10].

Importantly, animal models can be highly unreliable in the prediction of human responses due to interspecies variation relating to physiology, metabolism, and genetic makeup. Thus, animal

studies often do not translate to human use due to physiological differences, limiting their relevance and applicability [11]. Drug testing is particularly impacted by this, as most drugs identified to work in animals fail to translate to humans, with 89% of new drugs failing when entering clinical trials [11]. Taken together, this creates an urgent need for alternative models in muscle research with enhanced reliability and human relevance.

## 1.2 Skeletal muscle tissue engineering

Muscle tissue engineering holds great promise for the development of human in vitro models that could replace animals in research with numerous potential applications for drug development, drug screening, and the study of developmental or pathophysiological processes. Such models can, for example, be used to test novel pharmaceutical compounds prior to introducing them in humans to identify potential toxicities and determine efficacy in a human-relevant system.

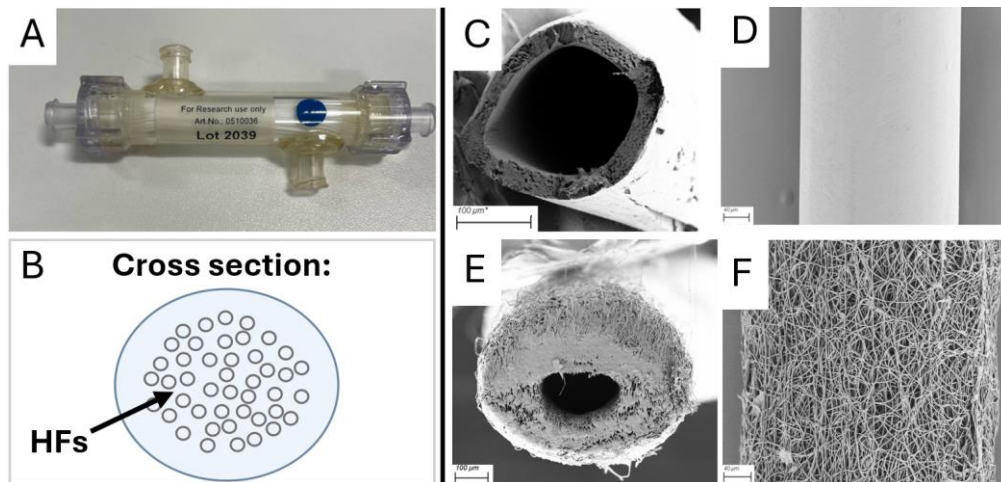
Despite the significant potential, the development of constructs that accurately mimic the complexity and heterogeneity of native tissues still presents a challenge today, which limits their clinical applicability and widespread adoption. One of the key challenges is the diffusion limit which impedes gas and nutrient exchange in thicker engineered tissues. Specifically, muscle has high metabolic demands and strongly depends on an efficient supply of oxygen and nutrients. Gases and nutrients can typically reach depths of 100-200 $\mu$ m by passive diffusion [12], with cells beyond this point undergo necrotic cell death. This largely restricts the size of in vitro tissues which limits their functionality and clinical applicability.

Another limiting factor is a lack of appropriate mechanical stimulation to support tissue development. Muscle tissue relies on mechanical input for growth and maturation as a highly dynamic tissue that undergoes cyclic loading and unloading during contraction and relaxation. Inside the body, different mechanical stresses act upon the tissue from multiple axes, which largely contributes to its effective development [13], [14]; however, traditional tissue engineering approaches insufficiently mimic the required mechanical stresses.

## 1.3 Hollow fibre bioreactor platforms

Hollow fibre (HF) bioreactors are used in tissue engineering and promise three-dimensional tissue culture with enhanced nutrient and oxygen transport (Fig. 2A). These modules contain semi-permeable HFs that provide a scaffold for growing tissue and mimic the function of the vasculature [15] (Fig. 2B). While providing enhanced nutrient and gas exchange, implementing HF modules that are soft and flexible could additionally enable mechanical stimulation through force transmission within a dynamic bioreactor system.

HFs are typically produced through extrusion, leading to smooth and stiff fibres that are not adapted to cell culture (Fig. 2C,D). Electrospun HFs (eHFs), on the other hand, are made up of microfibres and can be implemented to generate a flexible porous scaffold membrane for effective mechanical force transmission and diffusive exchange with the surrounding environment (Fig. 2E,F). Additionally, eHFs provide topographical stimuli, as their microfibres mimic the extracellular matrix (ECM) that surrounds cells inside the body to provide anchorage, structural signalling, and guidance cues [16], [17]. While eHF bioreactors show considerable promise, their scalability and functionality has yet to be demonstrated.



**Fig. 2: Hollow fibre bioreactor platforms**

**A.** Hollow fibre (HF) bioreactor module from AlphaPlan, **B.** Cross section of a HF bioreactor module (created with BioRender), **C.** Cross-sectional scanning electron microscopy (SEM) image of a commercial HF taken from the AlphaPlan bioreactor module, Scale bar: 100µm, **D.** Commercial HF imaged from above, Scale bar: 40µm, **E.** Cross-sectional SEM image of an electrospun hollow fibre (eHF), Scale bar: 100µm, **F.** eHF imaged from above with visible microfibres, Scale bar: 40µm.

## 1.4 Research objectives

The aim of this project is to develop a novel in vitro model of human skeletal muscle with enhanced physiological relevance and applicability in research and preclinical drug screening. By integrating the use of electrospun fibres, HF bioreactors, and dynamic platforms, key challenges in muscle tissue engineering, i.e. nutrient diffusion and mechanical stimulation, will be addressed. The experimental plan will involve 1. development of an eHF scaffold, 2. development of a soft flexible eHF bioreactor chamber, and 3. analysis of the effects of mechanical stimulation and pharmaceutical compounds on cultured muscle tissue.

### Objective 1: Developing an electrospun hollow fibre scaffold

The eHFs made of polycaprolactone (PCL) and polydioxanone (PDO) will be produced based on previous work [18]. In that work, a stainless-steel wire was used as a collector during electrospinning to generate eHFs. This required manual removal of individual wires post-spinning. For enhanced scalability and to facilitate integration of eHFs into the bioreactor chamber, the production and application of a dissolvable fibre collector will be investigated in this project. Based on this approach, hundreds of fibres could be embedded within the bioreactor chamber and the collector cores could be dissolved simultaneously.

The first challenge in this objective will be to develop a fibre collector with the following requirements:

- Conductive (max. 150MΩ resistance)
- Dissolvable
- Force-resistant
- Diameter: ~200µm

The second challenge will be to produce the eHFs with the following characteristics:

- Microfibre diameter: 0.5-1µm
- Wall thickness: 50µm



- Aligned topography

The desired microfibre thickness is 0.5-1 $\mu$ m given that collagen fibrils, which are highly abundant in skeletal muscle ECM, exhibit a diameter of up to 0.5 $\mu$ m [19]. Previous studies additionally found that electrospun microfibrils with a diameter of  $\sim$ 1 $\mu$ m provide topographical cues promote cell alignment [20]. A wall thickness of  $\sim$ 50 $\mu$ m is targeted to facilitate efficient gas and nutrient exchange while maintaining technical feasibility. For reference, the endothelium of human capillaries is up to 10 $\mu$ m thick [21]. Microfibrils will be aligned through stretching of the eHFs to incorporate topographical guidance which is known to support differentiation of myoblasts into myofibrils [22].

### **Objective 2: Developing a soft flexible eHF bioreactor chamber**

A soft flexible bioreactor chamber will be designed to provide synthetic vascularisation and transmission of mechanical signals. The eHF scaffold will be integrated into the chamber by fibre potting. Various analyses will be applied to ensure biocompatibility, efficient medium flow, and chamber and scaffold integrity under mechanical loading. More details can be found in the 'Future Work' section.

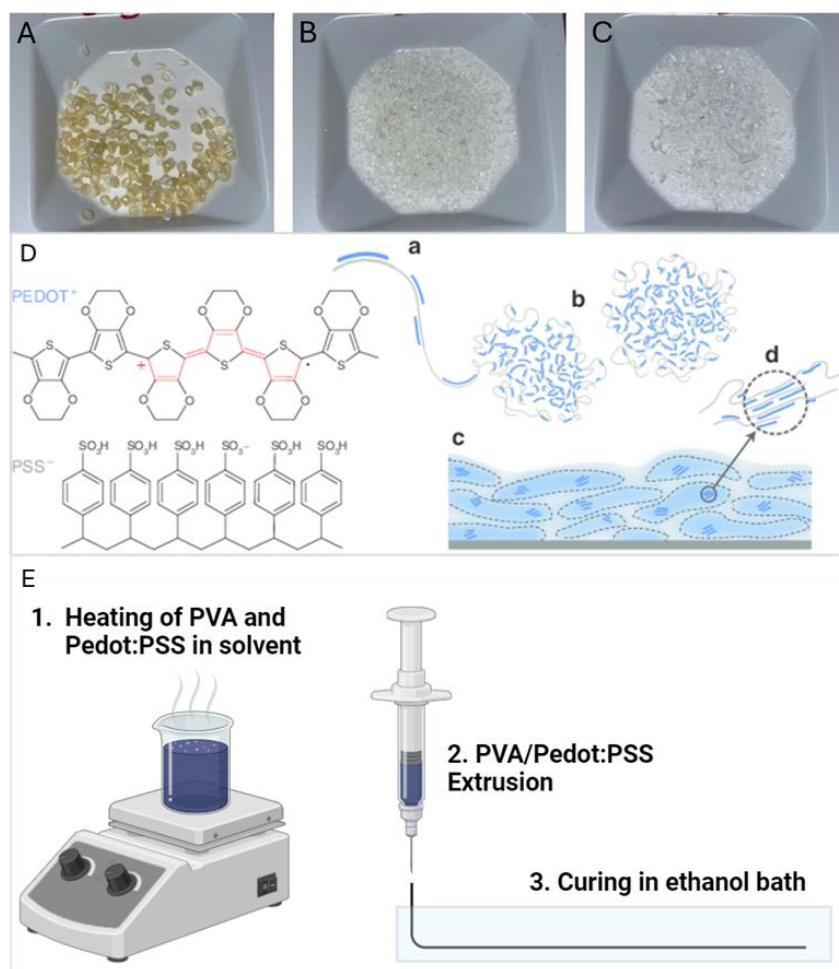
### **Objective 3: Applying the eHF bioreactor in skeletal muscle research**

Myoblasts will be seeded onto the eHFs inside the bioreactor chamber. Myogenesis will be induced, and the effects of mechanical loading will be investigated based on morphological features, protein expression, and functionality of the developed tissue. Additionally, the muscle's preliminary utility for drug testing will be assessed. More details can be found in the 'Future Work' section.

## **2. Materials and Methods**

### **2.1 Fabrication of a dissolvable fibre collector**

A protocol for a dissolvable electrospinning collector was established. The collector was synthesized using poly(vinyl alcohol) (PVA) as the base material, dimethyl sulfoxide (DMSO), and poly(3,4-ethylenedioxythiophene) polystyrene sulfonate (PEDOT:PSS) to provide conductive properties (Fig. 3D). PVA from three different source was tested: eSUN 14kDa Mw (3D Filament eSUN), Sigma 13-23kDa Mw (363170 Sigma Aldrich), Sigma 31-50kDa Mw (363073 Sigma Aldrich) (Fig. 3A,B,C). PVA was dissolved in water using sonication, and DMSO (D2650 Sigma Aldrich) and PEDOT:PSS (655201 Sigma Aldrich) were added to the solution. The resulting mixture was heated (95°C) under constant mechanical stirring (300rpm) to evaporate water. The mixture was then transferred to a syringe and extruded into a curing bath of ethanol absolute (UN1170 Sigma Aldrich) (Fig. 3E). A stainless-steel dispensing tip was used with an inner diameter of either 233 $\mu$ m (MT30-PBN Adhesive Dispensing Ltd) or 335 $\mu$ m (MT27-PBN Adhesive Dispensing Ltd). The resulting dissolvable fibre collector was collected and left overnight to allow for evaporation of ethanol and water. Some of the produced collectors were exposed to heat treatment at 37°C for 2 hours.



**Fig. 3: Production of the dissolvable fibre collector**

**A.** eSUN PVA filaments with 14kDa Mw, **B.** Sigma PVA crystals with 13-23kDa Mw, **C.** Sigma PVA crystals with 31-50kDa Mw, **D.** Molecular structure of PEDOT:PSS with PEDOT<sup>+</sup> core and PSS<sup>-</sup> shells [23], **E.** Schematic of the production of the dissolvable fibre collector (created with BioRender).

## 2.2 Characterisation of the dissolvable fibre collector

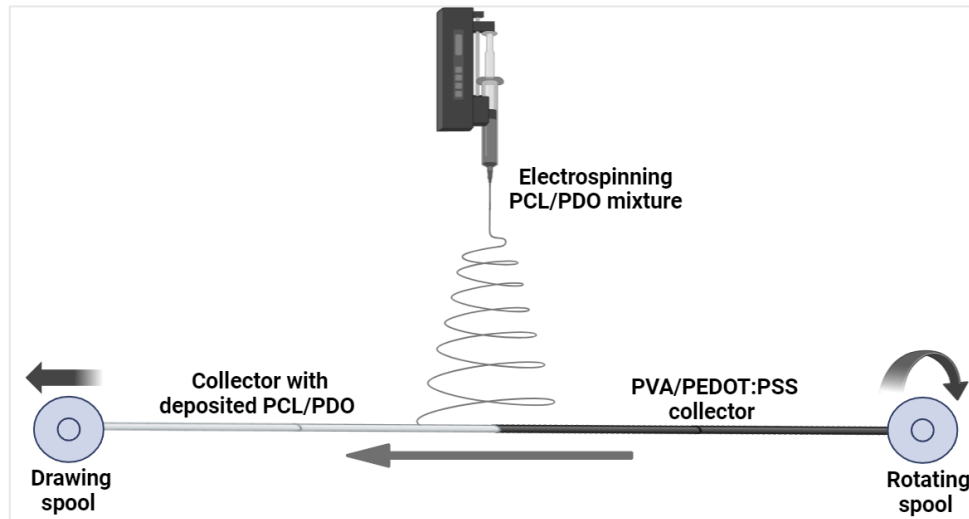
Conductivity of the fibre collector was determined based on resistance measurements of 10cm long collector sections using a Keithley 2400 Source Meter. Dissolvability was tested using 20cm long sections in 40mL water under constant stirring (240rpm) at room temperature (RT). Uniaxial tensile mechanical properties were tested using a ZwickRoell tensile tester (KAP-TC) with testXpert II at a speed of 10mm/min.

## 2.3 Fabrication of electrospun hollow fibres

Electrospinning solutions were prepared by mixing pyridine (110-86-1 Emsure Sigma Aldrich) into Hexafluoroisopropanol (HFIP; UN2922 Halocarbon) at a concentration of 50ppm. Polycaprolactone (PCL; Viatel C18E Ashland) and polydioxanone (PDO; RPM0400 Riverpoint Medical) were added to the HFIP/pyridine mixture at a ratio of 0.43:1 PCL to PDO, with resulting concentrations of 4.8% PCL and 11.1% PDO and a total polymer concentration (i.e. PCL+PDO) of 15.9%.

Electrospinning was performed using a rotating collector, where either a 200µm stainless-steel wire (SS316 The Crazy Wire Company) or the dissolvable fibre collector were used. The polymer solution was extruded at a speed of 0.8mL/hr with a distance of 19cm between tubing outlet and

collector. Collector rotation was set to 75rpm, and varying draw speeds of either 1mm/s, 1.5mm/s, or 2mm/s were used. A voltage of 9kV was applied. Following electrospinning, samples were stored overnight to allow for evaporation of residual HFIP. The stainless-steel wire was removed by manual pulling following soaking in 70% ethanol. The dissolvable collector was removed by pre-soaking in water at either RT or 37°C and subsequent five 5-minute sonication runs (7.5MJ, 40% amp) in water.



**Fig. 4: Production of electrospun hollow fibres**

The collector is drawn by a spool on the left and rotation is exerted by a spool on the right. PCL/PDO is electrospun and deposited onto the collector. Following electrospinning, the dissolvable core is removed, and the hollow fibres are stretched (created with BioRender).

## 2.4 Characterisation of electrospun hollow fibres

Light microscopy was used to determine the inner diameter, outer diameter, and wall thickness of eHFs and to detect potential changes in the fibre collector diameter as a result of plastic deformation sustained during electrospinning. Samples were imaged using ZEN Pro connected to an Olympus BX40 microscope with x10 magnification. Scanning electron microscopy (SEM) was used to assess eHF microfibre diameter and alignment. Samples were prepared with a Quorum sputter coater (Q150R ES) using QT Timed Gold at 20mA and 120s and imaged using Zeiss SmartSEM connected to a Zeiss scanning electron microscope (Sigma 300 FEG) with 80x, 220x, and 4.33kx magnification. Commercial HFs (AlphaPlan 0510036) composed of polymethylpentene with an inner diameter of 180µm, an outer diameter of 260µm, and a wall thickness of 80µm were used as a control.

## 2.5 Image analysis

Light microscopy and SEM images were analysed using Fiji ImageJ. Measurements of fibre collectors and eHFs were obtained using the distance measurement tool. The total diameter, wall thickness, and lumen diameter of eHFs were calculated based on detected inner and outer diameters. Measurements of microfibre diameters were obtained using the distance measurement tool. Microfibre alignment was assessed using the Fiji Directionality plugin. The derived data was used for quantitative analyses. Statistical significance is given as  $P \leq 0.05$  (\*),  $P \leq 0.01$  (\*\*),  $P \leq 0.001$  (\*\*\*), and  $P \leq 0.0001$  (\*\*\*\*).

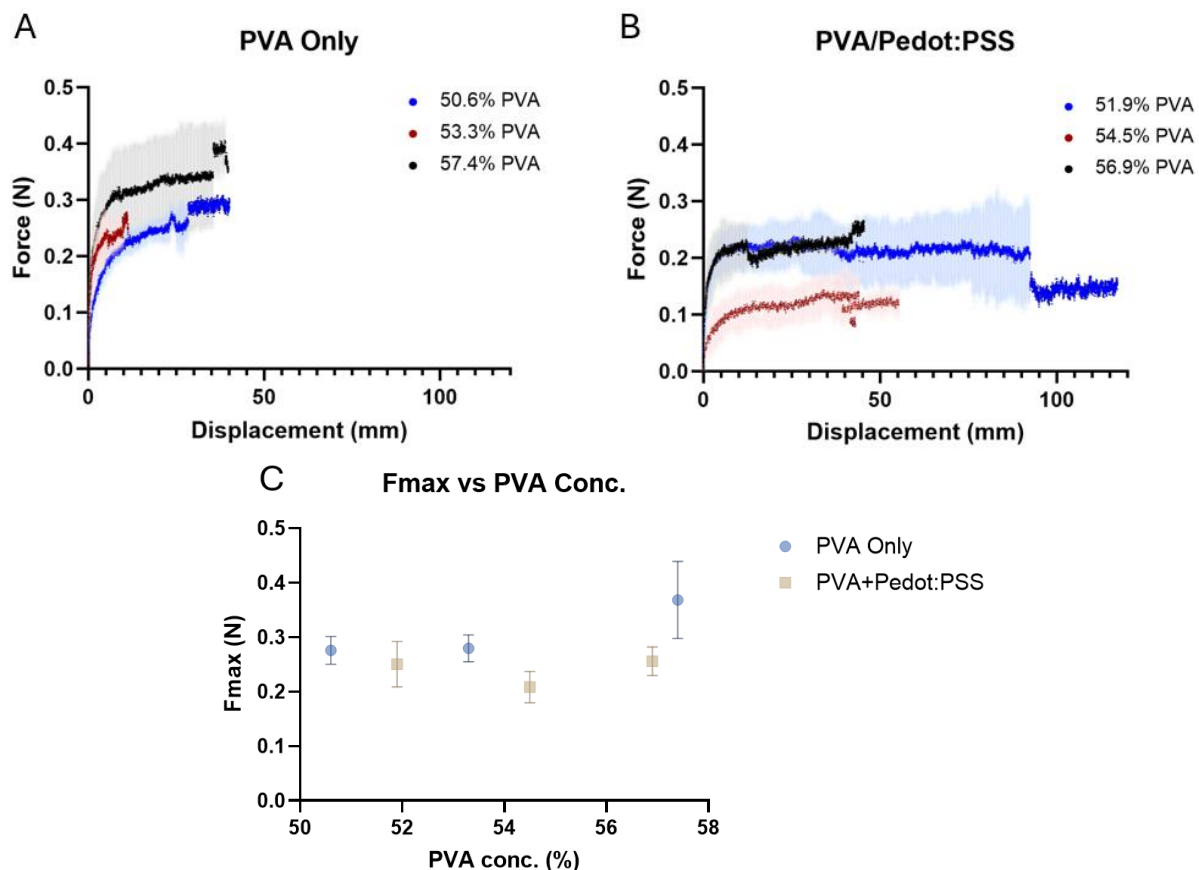
## 3. Results and Discussion

### 3.1 Production of a dissolvable fibre collector

#### 3.1.1 Effects of PVA concentration on wet-spinning and mechanical properties

PVA serves as the base material of the dissolvable fibre collector, and its concentration plays a crucial role in efficient extrusion and curing during wet-spinning. Early experiments focused on the use of PVA Only without PEDOT:PSS and, based on a curing test (Fig. S1), an initial range of 44-65%(w/v) PVA was selected for extrusion testing. The extrusion tests revealed that concentrations above 58% were too viscous to be extruded while concentrations below 50% led to discontinuous extrusion and adhesivity of the cured product. The introduction of PEDOT:PSS into the PVA solutions resulted in increased viscosities of the mixtures, thus favouring extrusion with PVA concentrations below 52%. Notably, it was observed that higher PEDOT:PSS concentrations necessitated lower PVA concentrations in order to maintain efficient extrusion and curing.

The mechanical properties of fibre collectors made from either PVA Only or PVA/PEDOT:PSS at PVA concentrations within the range of 50-58% were tested for comparison (Fig. 5A,B). Force-displacement curves showed a significantly lower travel distance for PVA Only, indicating lower levels of plastic deformation compared to PVA/PEDOT:PSS collectors; though, no significant difference could be detected in the maximum force sustained by the different collectors (Fig. 5C).



**Fig. 5: Tensile mechanical properties of the dissolvable fibre collector**

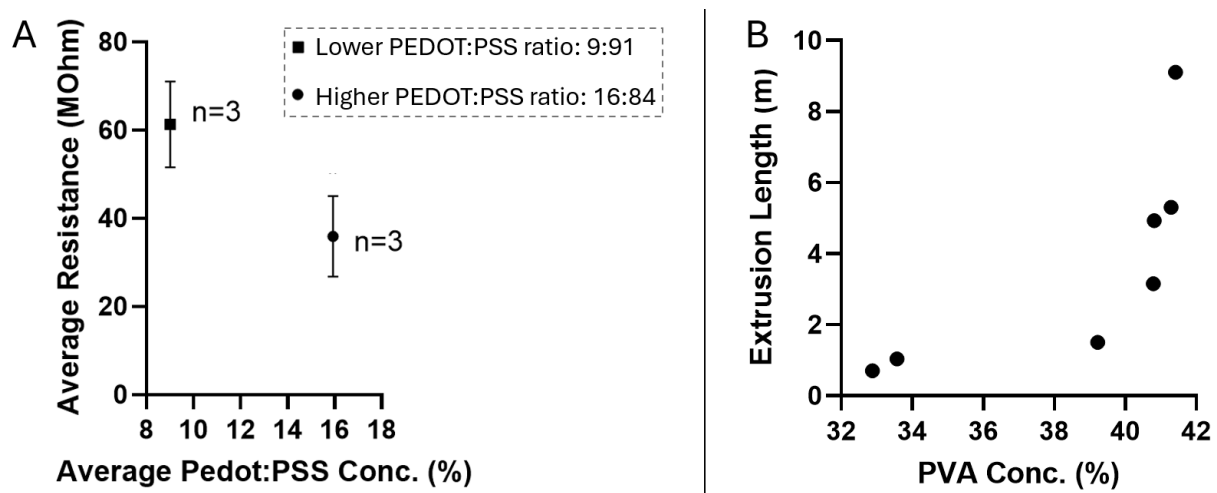
**A.** Force-displacement curve of PVA Only collectors with different PVA concentrations, **B.** Force-

displacement curve of dissolvable PVA/PEDOT:PSS collectors with different PVA concentrations, **C.** Fmax (N) over PVA concentration (%) of PVA Only and PVA/PEDOT:PSS collectors.

### 3.1.2 Achieving electrical conductivity and large extrusion length

Electrical conductivity of the fibre collector was crucial to enable the electrospinning process, as it must be grounded to attract electrospun fibres. The collector's conductivity is largely governed by the ratio of PEDOT:PSS-to-PVA content. Initial attempts using a PEDOT:PSS-to-PVA ratio of 4:96 yielded fibre collectors for which electrical resistance lay above the equipment's upper limit (200MΩ), suggesting poor conductivity. The PEDOT:PSS content was therefore increased, and ratios of 9:91 and 16:84 led to average resistances of  $61 \pm 10 \text{ M}\Omega$  and  $36 \pm 9 \text{ M}\Omega$ , respectively (Fig. 6A). Considering the costs associated with PEDOT:PSS, the lowest ratio (9:91) was selected for subsequent experiments to optimise for economic feasibility and scalability. The produced collectors were then tested in the electrospinner, and deposition of electrospun fibres was observed, which proved adequate levels of conductivity.

To ensure sufficient extrusion lengths, the PVA concentration for the 9:91 ratio had to be refined, and ~41% PVA allowed for the best outcomes (Fig. 6B). The final protocol was tested and confirmed to allow for the reproducible production of fibre collectors by applying a 9:91 PEDOT:PSS-to-PVA ratio with 41% PVA and 4.1% PEDOT:PSS.



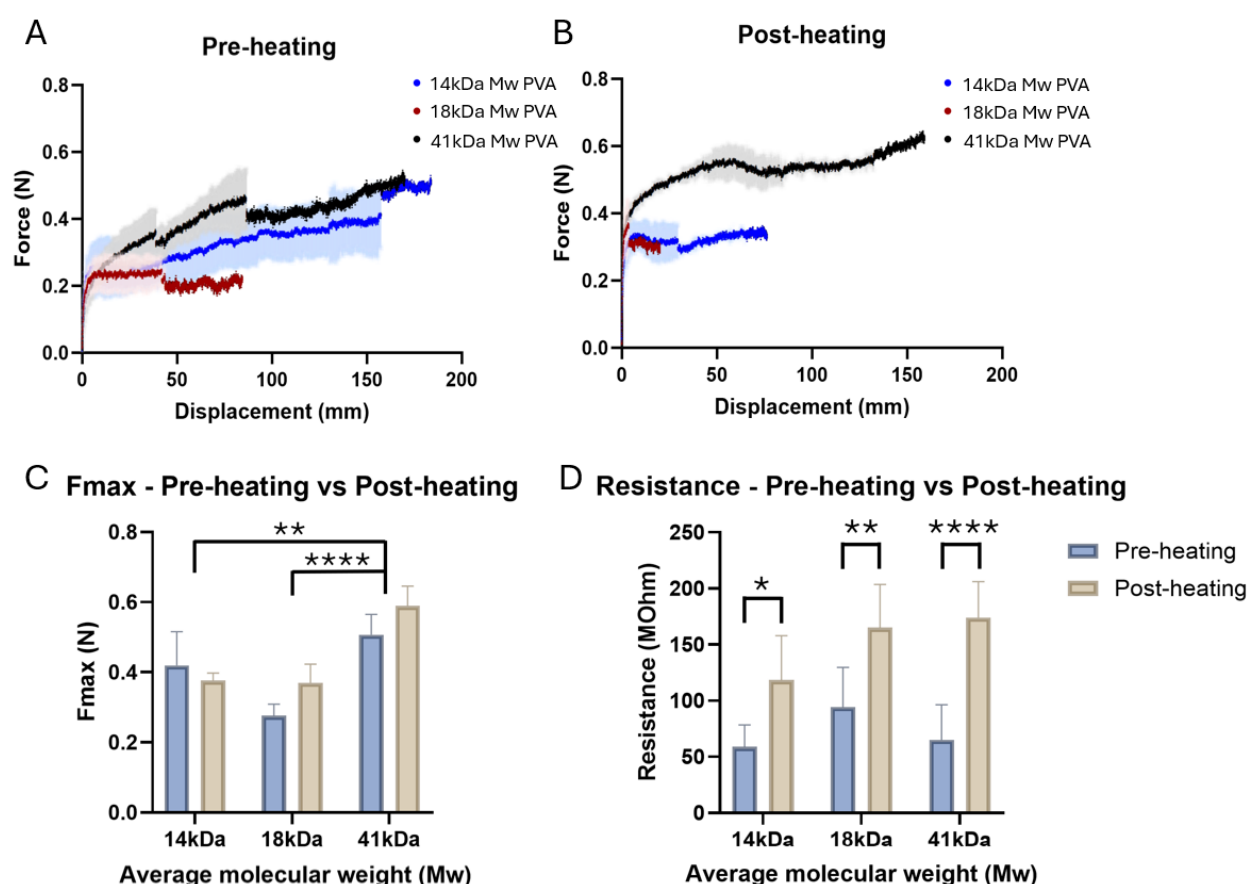
**Fig. 6: Effects of PEDOT:PSS and PVA concentrations on electrical resistance and extrusion length**  
**A.** Average resistance of dissolvable fibre collectors with PEDOT:PSS to PVA ratios of 16:84 and 9:91, **B.** Extrusion length (m) over PVA concentration (%).

### 3.1.3 Effects of PVA type and heating on fibre collector properties

Three types of PVA were tested for the production of fibre collectors, including PVA with average molecular weights (Mw) of 14kDa, 18kDa, and 41kDa. All collectors were produced with a PEDOT:PSS-to-PVA ratio of 9:91. Heat treatment (30°C, 2hrs) was applied to identify whether the collectors' strength could be increased.

Tensile tests were conducted to compare pre- and post-heating collectors made from different PVA types. The force displacement curves revealed reduced plastic deformation of 14kDa and 18kDa fibre collectors following heating while 41kDa collectors maintained their range (Fig. 7A,B). Heating did not significantly affect the maximum force (Fmax) of the collectors; however, 41kDa collectors showed a significantly higher Fmax than their 14kDa and 18kDa counterparts

(Fig. 7C). Post-heating 41kDa fibre collectors exhibited the highest force resistance. Electrical resistance was found to be lower for 14kDa samples compared to 18kDa and 41kDa ones (Fig. 7D), and heat treatment caused a significant increase in resistance, such that conductivity could not be detected for some sections of the 18kDa and 41kDa collectors. 14kDa samples maintained the lowest average resistance, measuring  $59\pm20\text{M}\Omega\text{m}$  pre-heating and  $118\pm40\text{M}\Omega\text{m}$  post-heating. Notably, the dissolvability test showed no significant differences between pre- or post-heating samples made from different PVA types, as all collectors dissolved effectively after six minutes of continuous stirring in water (Fig. S3).



**Fig. 7: Characterisation of pre- and post-heating fibre collectors made from different PVA types**

**A.** Force-displacement curve of different PVA types pre-heating, **B.** Force-displacement curve of different PVA types post-heating, **C.** Fmax (N) of fibre collectors made from different PVA types pre- and post-heating, **D.** Electrical resistance (MΩm) of fibre collectors made from different PVA types pre- and post-heating.

Despite elevated electrical resistance, it was decided to incorporate a post-spinning heating step for future work, as this improved the fibre collectors' stiffness, with an increase in Young's modulus from 3.9MPa to 31.5MPa. This is favourable, as it prevents excessive deformation of the fibre collector resulting from tension during the electrospinning process. While 41kDa collectors exhibited the highest force resistance, 14kDa collectors showed the lowest level of electrical resistance, with a post-heating average of  $120\pm40\text{M}\Omega\text{m}$  and a maximum of  $150\text{M}\Omega\text{m}$ . Given that collectors with  $R>150\text{M}\Omega\text{m}$  were observed to impair electrospun fibre deposition, 14kDa collectors appear to be the most suitable for electrospinning. Additionally, processing of the 14kDa PVA, which is provided in the form of filaments, is more favourable compared to the 18kDa and 41kDa PVA which are supplied as crystals that easily escape dissolution and thus result in



blockages during wet-spinning (Fig. 3). Based on these results, 14kDa PVA was selected for the subsequent steps of this work.

## 3.2 Production of electrospun hollow fibres

### 3.2.1 Suitability of the dissolvable fibre collector for electrospinning

After using a stainless-steel collector to identify the appropriate electrospinning parameters, such as tube-to-collector distance, solution extrusion speed, and voltage, the dissolvable fibre collector was tested for electrospinning of PCL/PDO eHFs. During this process, thinning of the dissolvable collector was observed due to tension generated by drawing it through the electrospinner. Additionally, it was found that after some time the PCL/PDO fibres only deposited onto one side of the collector, indicating insufficient rotation. Force transmission was tested by measuring the draw speed of a stainless-steel collector and a dissolvable fibre collector with an input speed of 0.5mm/s. While the stainless-steel wire showed speeds close to input ( $0.50\pm0.02\text{mm/s}$ ), the dissolvable collector exhibited significantly impaired force transmission ( $0.21\text{mm/s}$ ) and underwent mechanical failure. To address this, a tension controller and a tube for reduced friction during unwinding were installed (Fig. S4). When using this setup, the collector's integrity was maintained without failure or visible deformation and the same speed for both collectors ( $0.50\pm0.02\text{mm/s}$ ,  $0.50\pm0.01\text{mm/s}$ ) was recorded, indicating efficient force transmission.

The dissolvable fibre collector was then successfully applied for collecting PCL/PDO eHFs, as deposited material showed an even distribution, indicating functional rotation and sufficient electrical conductivity. Following electrospinning, measurements of the dissolvable collector diameter revealed that plastic deformation during electrospinning occurred within a reasonable range, with an average diameter reduction of  $5.3\pm4.5\%$  and a maximum of 10%. Finally, it was concluded that the produced dissolvable fibre collector presents a viable alternative to traditional stainless-steel collectors while enhancing scalability prospects.

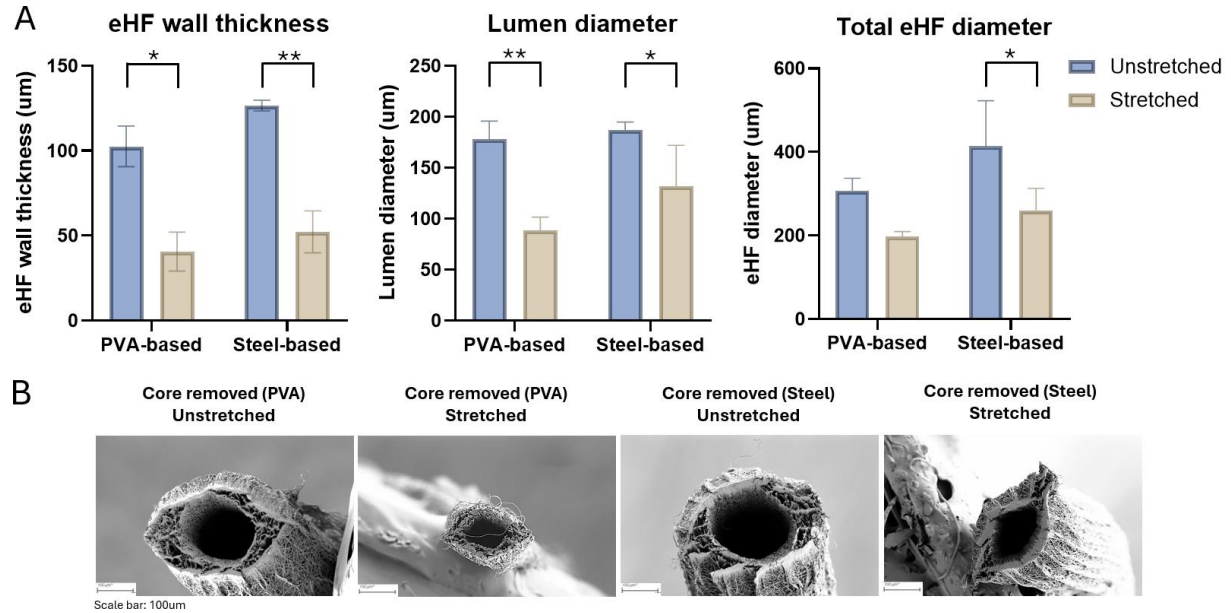
### 3.2.2 Effects of draw speed on electrospun hollow fibre wall thickness

The speed at which a collector is drawn through the electrospinner dictates how much material is deposited. This in turn affects the wall thickness of the produced eHFs. To identify which draw speed correlates with the desired dimensions, speeds of 1mm/s, 1.5mm/s, and 2mm/s were tested. Given that a final post-stretch wall thickness of  $\sim 50\mu\text{m}$  is desired and assuming a  $\sim 50\%$  reduction in wall thickness based on stretching, the eHFs were aimed to have a pre-stretch diameter of  $\sim 100\mu\text{m}$ . Based on the obtained light imaging data, a draw speed of 1mm/s correlated with a wall thickness of  $\sim 100\mu\text{m}$  (Fig. S5), with eHFs generated using the dissolvable fibre collector (PVA-based) exhibiting an average thickness of  $105\pm22\mu\text{m}$  and eHFs generated using the stainless-steel collector (steel-based) showing an average thickness of  $108\pm8\mu\text{m}$ . Based on the given results, a collector draw speed of 1mm/s was chosen for eHFs production.

### 3.2.3 Effects of stretching on electrospun hollow fibre dimensions

With the aim to establish longitudinally aligned microfibres for cell guidance, eHFs were stretched manually until resistance to plastic deformation was noted. The stretched PVA-based and steel-based eHFs showed an average length increase of  $2.6\pm0.3\times$  and  $2.3\pm0.4\times$ , respectively. Steel-based eHFs thus exhibited a lower increase in length, though no significant difference ( $p=0.3$ ) was detected. To further analyse the effects of stretching, eHFs were imaged by SEM. Post-stretch eHFs showed a clear structural alignment compared to pre-stretch eHFs (Fig. S8).

Significant reductions in eHF diameter, wall thickness, and lumen diameter were noted, with an overall diameter of  $198 \pm 12 \mu\text{m}$  and  $260 \pm 54 \mu\text{m}$ , wall thickness of  $41 \pm 11 \mu\text{m}$  and  $52 \pm 12 \mu\text{m}$ , and lumen diameter of  $89 \pm 13 \mu\text{m}$  and  $132 \pm 40 \mu\text{m}$  observed for PVA- and steel-based eHFs, respectively (Fig. 8A). The post-stretch wall thickness, therefore, aligned well with the desired dimensions. It may also be noted that eHFs with an intact or removed collector core exhibited no significant differences, indicating that neither manual removal of the steel collector nor flushing of the dissolvable collector affect fibre quality.



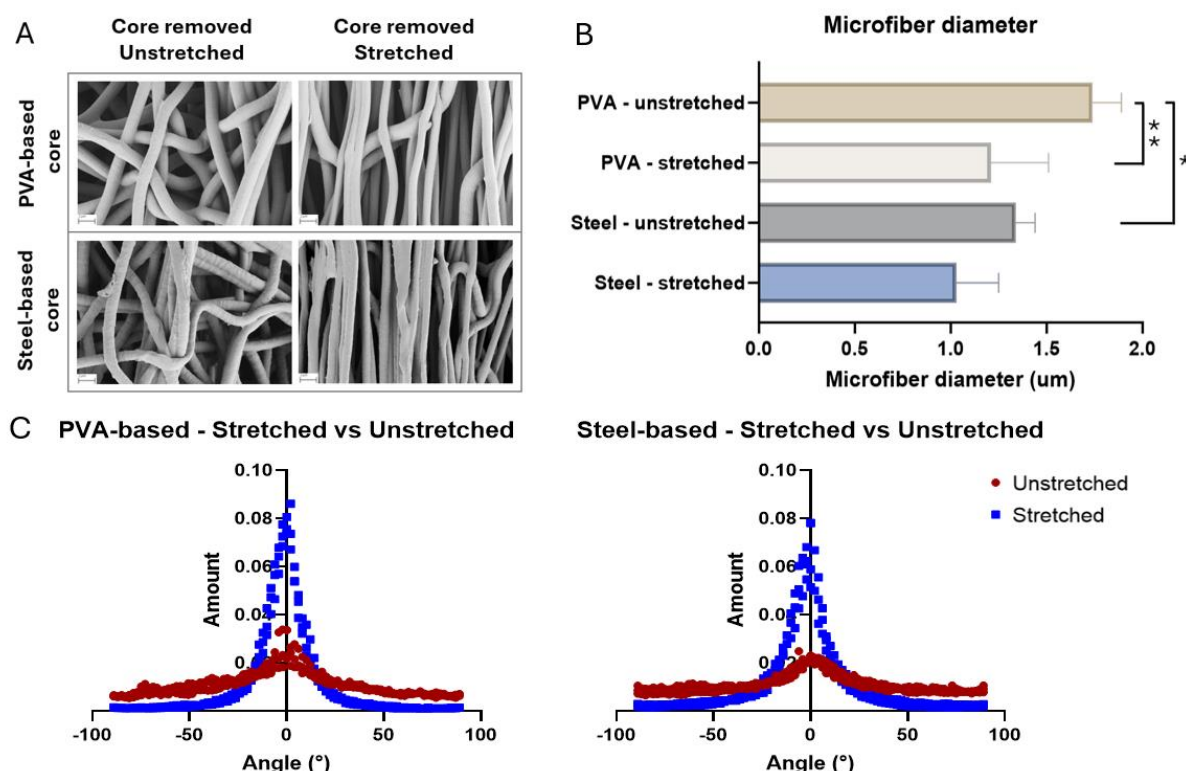
**Fig. 8: Dimensions of pre- and post-stretch electrospun hollow fibres**

**A.** Hollow fibre (HF) wall thickness, lumen diameter, and HF diameter of unstretched and stretched PVA- and steel-based electrospun HFs, **B.** Cross-sectional scanning electron microscopy (SEM) images at a magnification of 160x, Scale bars: 100μm.

### 3.2.4 Effects of stretching on microfibre diameter and alignment

SEM images at high magnification were used to analyse the microfibrils of which the eHFs are composed. A 30% and 23% reduction in microfibre diameter was observed following stretching, leading to diameters of  $1.21 \pm 0.30 \mu\text{m}$  for PVA-based and  $1.03 \pm 0.22 \mu\text{m}$  for steel-based eHFs, respectively (Fig. 9B). Given a desired post-stretch microfibre diameter of 0.5-1μm, this aligns with the targeted dimensions. The unstretched microfibre diameter of PVA-based eHFs was slightly larger ( $+0.4 \mu\text{m}$ ;  $p=0.04$ ) than that of steel-based eHFs. This could potentially be explained by the higher electrical conductivity of stainless steel compared to PVA/PEDOT:PSS. A significant difference ( $p=0.006$ ) was detected between the microfibre diameter of pre- and post-stretch PVA-based eHFs. Despite a clear reduction in microfibre diameter from  $1.34 \pm 0.10 \mu\text{m}$  pre-stretch to  $1.03 \pm 0.22 \mu\text{m}$  post-stretch, significance could not be determined for steel-based eHFs. No significant differences were observed between eHFs pre- and post-core removal. Alignment of eHFs presents an essential aspect, as it provides topographical guidance to aid the arrangement and differentiation of cells. Post-stretch microfibrils showed a clear structural alignment (Fig. 9A), and directionality histograms displayed a significant difference ( $p<0.0001$ ) pre- and post-stretch, as post-stretch microfibrils exhibited increased alignment along the 0° axis (Fig. 9C). No significant differences were observed between pre- and post-core removal eHFs.





**Fig. 9: Diameter and alignment of pre- and post-stretch electrospun microfibres**

**A.** Scanning electron microscopy (SEM) images showing microfibres of electrospun hollow fibres (eHF) at a magnification of 4.33k, Scale bars: 2μm, **B.** Microfibre diameter (μm) of unstretched and stretched PVA- and steel-based eHFs, **C.** Directionality histograms of unstretched and stretched PVA- and steel-based eHFs.

### 3.2.5 Removal of the dissolvable collector core

Following generation of the eHFs, the dissolvable collector needed to be removed from the eHF core by subjecting them to sonication. Fibre sections were placed in water and sonicated with five 5-minute runs at an amplitude of 40%. Following a total sonication time of 25 minutes, the collector had not dissolved (Fig. S10). To address this, fibre sections were pre-soaked in water at RT (min. 8h) prior to sonication; however, following five 5-minute sonication runs at 40% amplitude, significant quantities of collector residue were still visible. Next, eHFs were pre-soaked in 37°C (min. 8h) which resulted in clearly enhanced collector removal, indicating that elevated soaking temperatures facilitate the process. Nonetheless, some of the collector material remained. Applying two rounds of sonication following pre-soaking at RT plus pre-soaking at 37°C did not alleviate this. Full removal of the collector material by perfusion will be tested in the future.

## 3.3 Limitations

Several limitations related to methodology and materials need to be noted. Methodological limitations include the following. Experimenter's experience influences wet-spinning and electrospinning outcomes. Wet-spinning, collector core removal and eHF stretching are manual processes which can introduce errors and variability. Viscosity measurements of the produced mixtures were not obtained. Materials limitations include the following. Only a limited molecular weight range for PVA was tested. The produced dissolvable collector exhibited moisture sensitivity, with excessive moisture loss leading to brittleness. A slight deterioration in the quality of eHFs was observed based on their morphological features, which could be linked to the use of

PDO that in its unprocessed form degrades over the course of a few months. Finally, complete removal of the dissolvable collector core of eHF was not yet achieved.

## 4. Conclusions and Future Work

### 4.1 Conclusions

A protocol for the production of a reproducible dissolvable electrospinning collector was established. This collector was successfully employed to generate eHFs. The dissolvable fibre collector performed well compared to a traditional steel collector, and the removal process showed to have no significant effects on fibre quality. Electrospinning parameters required to achieve the desired fibre dimensions were identified. Furthermore, stretch-induced alignment of microfibres was confirmed. The obtained results set a good foundation for future work that will explore application of the generated eHF scaffold inside a bioreactor module to address current limitations in the field of skeletal muscle tissue engineering.

### 4.2 Future Work

#### 4.2.1 Electrospun hollow fibre scaffold

The possibility of eliminating PDO from the current protocol will be explored, and removal of the dissolvable collector will be further investigated to ensure complete flushing of the collector material. Additionally, various analyses will be applied to further characterise the produced eHF scaffold. For example, micro-computed tomography (micro-CT) will be used to determine the scaffold's pore size, porosity percentage, and potential spatial variations in porosity. Uniaxial tensile mechanical testing will be used to determine the mechanical properties of the eHFs. Biocompatibility of the scaffold will be tested based on a cytotoxicity assay. The structural and biological support the scaffold offers to cells will be analysed by assessing attachment, morphology, proliferation, and differentiation of cultured cells. Furthermore, differentiation of cells cultured on stretched versus unstretched eHFs will be assessed.

#### 4.2.2 The bioreactor chamber and mechanical loading

The developed eHF scaffold is aimed to be incorporated into a soft flexible bioreactor chamber designed for the integration into dynamic mechanical platforms such as uniaxial stages or more complex multiaxial systems, e.g. the humanoid bioreactor platform developed by our group [24]. The chamber will be designed to host the eHF scaffold, support the influx and outflux of culture medium, and allow for efficient mechanical signal transduction. The final design is aimed to accommodate several hundred eHFs with direct perfusion via ports. A direct inlet and outlet to the chamber will additionally allow for medium changes with muscle-specific differentiation factors. The developed chamber will be assessed for its functionality and integrity, specifically during fluid flow and mechanical loading. Once functionality has been confirmed, eHFs with intact dissolvable collector cores will be plotted into the chamber followed by simultaneous core flushing. Subsequently, a uniaxial tension device will be tested for uniform stretching of all eHFs. Effective force transmission within the eHF bioreactor module will need to be ensured and stability of the scaffold under mechanical loading must be confirmed. Human myoblasts will then be seeded onto the scaffold inside the chamber followed by mechanical stimulation.

### 4.2.3 Functionality and utility of the muscle tissue construct

Typical morphological features of muscle tissue will be assessed, including sarcomeric banding patterns, multinucleation, and a hierarchical tubular architecture [25]. Myotube alignment will be measured based on directionality assays, and the myoblast fusion index will be identified based on the number of detected nuclei per myotube. Notably, increased levels of alignment and a higher fusion index correlate with enhanced tissue maturity [26]. Functional features such as twitching and contractile force generation will also be evaluated. Using immunofluorescent staining and quantitative polymerase chain reaction (qPCR), the expression levels of skeletal muscle differentiation markers will be identified to determine the stages of tissue maturation. This will likely include myoblast determination protein 1 (MyoD), myogenin, and myosin heavy chain (MHC), which are linked to proliferation, differentiation, and myotube maturation, respectively [27]. Other skeletal muscle differentiation markers that may be tested include alpha-actinin and desmin [28].

Once synthesis of mature muscle has been confirmed, preliminary drug response assays will be performed to analyse responses to drugs with well-known effects, such as blebbistatin, a myosin II inhibitor known to suppresses muscle contraction [29], and chemotherapy drug cisplatin, a toxic agent known to induce a decrease in muscle mass and myofibre diameter causing skeletal muscle dysfunction [30].

#### 4.2.4 Gantt Chart

|                                                 | AY2024/25 |         |        | AY2025/26  |        |         | AY2026/27 |            |        |         |        |
|-------------------------------------------------|-----------|---------|--------|------------|--------|---------|-----------|------------|--------|---------|--------|
|                                                 | Hilary    | Trinity | Summer | Michaelmas | Hilary | Trinity | Summer    | Michaelmas | Hilary | Trinity | Summer |
| Development of hollow fibre scaffold            |           |         |        |            |        |         |           |            |        |         |        |
|                                                 |           |         |        |            |        |         |           |            |        |         |        |
|                                                 |           |         |        |            |        |         |           |            |        |         |        |
|                                                 |           |         |        |            |        |         |           |            |        |         |        |
| Development of soft flexible bioreactor chamber |           |         |        |            |        |         |           |            |        |         |        |
|                                                 |           |         |        |            |        |         |           |            |        |         |        |
|                                                 |           |         |        |            |        |         |           |            |        |         |        |
|                                                 |           |         |        |            |        |         |           |            |        |         |        |
| Effects of mechanical loading & Drug response   |           |         |        |            |        |         |           |            |        |         |        |
|                                                 |           |         |        |            |        |         |           |            |        |         |        |
|                                                 |           |         |        |            |        |         |           |            |        |         |        |
|                                                 |           |         |        |            |        |         |           |            |        |         |        |
| Thesis Write-Up & Publications                  |           |         |        |            |        |         |           |            |        |         |        |
|                                                 |           |         |        |            |        |         |           |            |        |         |        |
|                                                 |           |         |        |            |        |         |           |            |        |         |        |
|                                                 |           |         |        |            |        |         |           |            |        |         |        |

#### 4.2.5 Risk mitigation plan

|                                     | Risk                                     | Mitigation                                       | Contingency                                    |
|-------------------------------------|------------------------------------------|--------------------------------------------------|------------------------------------------------|
| <b>eHF Scaffold</b>                 | Failure to produce dissolvable collector | Training by group's materials engineer           | Use stainless-steel collector                  |
|                                     | Electrospinner failure                   | Training for proper handling                     | Alternative device at engineering              |
|                                     | No cell attachment                       | Training by group's biologist                    | Coating for anchorage points                   |
| <b>Bioreactor Chamber</b>           | eHFs loosen inside chamber               | Training by group's materials engineer           | Improve potting method                         |
|                                     | Failure of chamber function              | Experience in the group with preliminary work    | Circumvent use of a soft membrane              |
| <b>Tissue growth &amp; analysis</b> | Impaired formation of mature muscle      | Training by group's biologist                    | Adjust differentiation protocol                |
|                                     | Contamination                            | Training by group's biologist                    | Use different cells                            |
|                                     | No response to drugs                     | Expertise of group's biologist and co-supervisor | Enhance tissue maturity<br>Try different drugs |

## 5. References

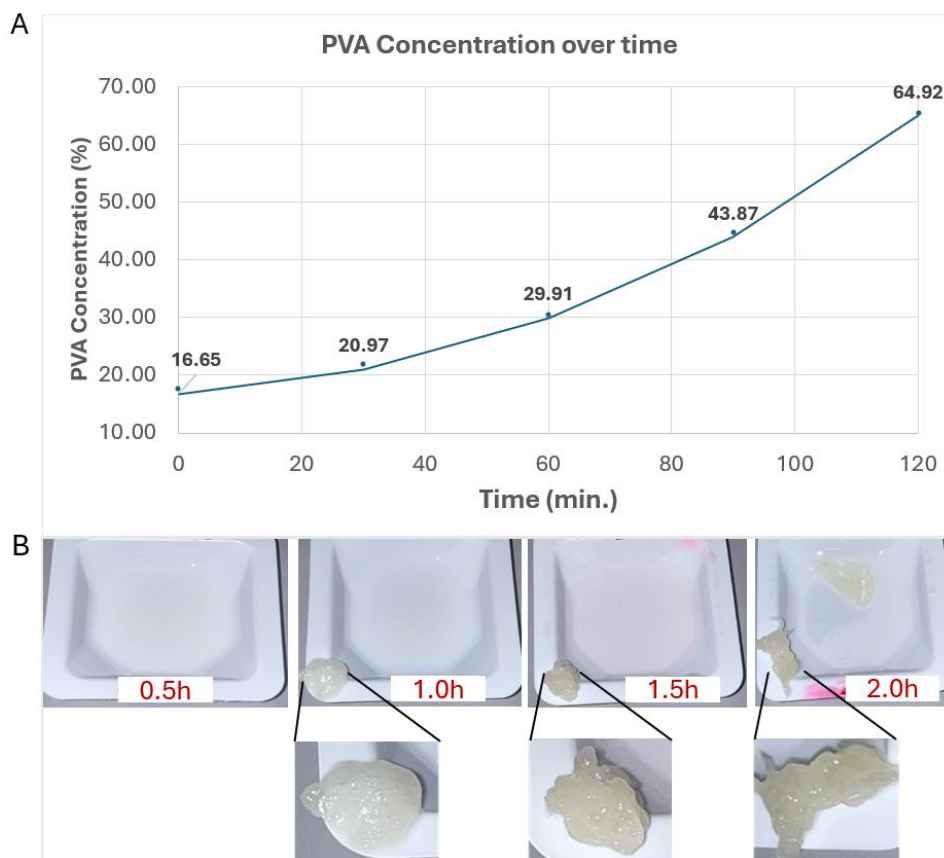
- [1] C. McCuller, R. Jessu, and A. L. Callahan, 'Physiology, Skeletal Muscle', in *StatPearls*, Treasure Island (FL): StatPearls Publishing, 2024. Accessed: Dec. 02, 2024. [Online]. Available: <http://www.ncbi.nlm.nih.gov/books/NBK537139/>
- [2] T. Zidarič, M. Milojević, J. Vajda, B. Vihar, and U. Maver, 'Cultured Meat: Meat Industry Hand in Hand with Biomedical Production Methods', *Food Eng. Rev.*, vol. 12, no. 4, pp. 498–519, Dec. 2020, doi: 10.1007/s12393-020-09253-w.
- [3] 'Skeletal Muscle | Anatomy and Physiology I'. Accessed: Dec. 09, 2024. [Online]. Available: <https://courses.lumenlearning.com/suny-ap1/chapter/skeletal-muscle/>
- [4] A. Kaczmarek *et al.*, 'The Role of Satellite Cells in Skeletal Muscle Regeneration—The Effect of Exercise and Age', *Biology*, vol. 10, no. 10, p. 1056, Oct. 2021, doi: 10.3390/biology10101056.
- [5] 'Musculoskeletal health'. Accessed: Dec. 02, 2024. [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/musculoskeletal-conditions>
- [6] E. M. McNally and P. Pytel, 'Muscle Diseases: The Muscular Dystrophies', *Annu. Rev. Pathol. Mech. Dis.*, vol. 2, no. Volume 2, 2007, pp. 87–109, Feb. 2007, doi: 10.1146/annurev.pathol.2.010506.091936.
- [7] R. R. Wolfe, 'The underappreciated role of muscle in health and disease<sup>2</sup>', *Am. J. Clin. Nutr.*, vol. 84, no. 3, pp. 475–482, Dec. 2006, doi: 10.1093/ajcn/84.3.475.
- [8] 'ALURES statistics - Section 2 : Uses of animals for research, testing, routine production and education and training purposes - European Commission', Environment Alures statistics. Accessed: Dec. 02, 2024. [Online]. Available: [https://webgate.ec.europa.eu/envdataportal/content/alures/section2\\_number-of-uses.html](https://webgate.ec.europa.eu/envdataportal/content/alures/section2_number-of-uses.html)
- [9] V. T. von Kortzfleisch *et al.*, 'Do multiple experimenters improve the reproducibility of animal studies?', *PLOS Biol.*, vol. 20, no. 5, p. e3001564, May 2022, doi: 10.1371/journal.pbio.3001564.
- [10] C. B. Brink and D. I. Lewis, 'The 12 Rs Framework as a Comprehensive, Unifying Construct for Principles Guiding Animal Research Ethics', *Animals*, vol. 13, no. 7, Art. no. 7, Jan. 2023, doi: 10.3390/ani13071128.
- [11] G. A. Van Norman, 'Limitations of Animal Studies for Predicting Toxicity in Clinical Trials: Is it Time to Rethink Our Current Approach?', *JACC Basic Transl. Sci.*, vol. 4, no. 7, pp. 845–854, Nov. 2019, doi: 10.1016/j.jacbts.2019.10.008.
- [12] T. Rademakers, J. M. Horvath, C. A. van Blitterswijk, and V. L. S. LaPointe, 'Oxygen and nutrient delivery in tissue engineering: Approaches to graft vascularization', *J. Tissue Eng. Regen. Med.*, vol. 13, no. 10, pp. 1815–1829, 2019, doi: 10.1002/term.2932.
- [13] C. A. Powell, B. L. Smiley, J. Mills, and H. H. Vandenberg, 'Mechanical stimulation improves tissue-engineered human skeletal muscle', *Am. J. Physiol.-Cell Physiol.*, vol. 283, no. 5, pp. C1557–C1565, Nov. 2002, doi: 10.1152/ajpcell.00595.2001.

- [14] Y. Wang *et al.*, 'Multiple Effects of Mechanical Stretch on Myogenic Progenitor Cells', *Stem Cells Dev.*, Mar. 2020, doi: 10.1089/scd.2019.0286.
- [15] H. Eghbali, M. M. Nava, D. Mohebbi-Kalhari, and M. T. Raimondi, 'Hollow Fiber Bioreactor Technology for Tissue Engineering Applications', *Int. J. Artif. Organs*, vol. 39, no. 1, pp. 1–15, Jan. 2016, doi: 10.5301/ijao.5000466.
- [16] J. Nicolas, S. Magli, L. Rabbachin, S. Sampaolesi, F. Nicotra, and L. Russo, '3D Extracellular Matrix Mimics: Fundamental Concepts and Role of Materials Chemistry to Influence Stem Cell Fate', *Biomacromolecules*, vol. 21, no. 6, pp. 1968–1994, Jun. 2020, doi: 10.1021/acs.biomac.0c00045.
- [17] S. Ferraris *et al.*, 'Topographical and Biomechanical Guidance of Electrospun Fibers for Biomedical Applications', *Polymers*, vol. 12, no. 12, Art. no. 12, Dec. 2020, doi: 10.3390/polym12122896.
- [18] R. J. Martin, 'Development of vascular inspired electrospun hollow fibre membranes for use in 3D tissue culture', <http://purl.org/dc/dcmitype/Text>, University of Oxford, 2022. Accessed: Dec. 09, 2024. [Online]. Available: <https://ora.ox.ac.uk/objects/uuid:96e65892-b8d3-41c3-a50c-29819e43eb84>
- [19] S. M. Siadat, A. A. Silverman, C. A. DiMarzio, and J. W. Ruberti, 'Measuring Collagen Fibril Diameter with Differential Interference Contrast Microscopy', *J. Struct. Biol.*, vol. 213, no. 1, p. 107697, Mar. 2021, doi: 10.1016/j.jsb.2021.107697.
- [20] S. Y. Chew, R. Mi, A. Hoke, and K. W. Leong, 'The effect of the alignment of electrospun fibrous scaffolds on Schwann cell maturation', *Biomaterials*, vol. 29, no. 6, pp. 653–661, Feb. 2008, doi: 10.1016/j.biomaterials.2007.10.025.
- [21] M. Félétou, 'Introduction', in *The Endothelium: Part 1: Multiple Functions of the Endothelial Cells—Focus on Endothelium-Derived Vasoactive Mediators*, Morgan & Claypool Life Sciences, 2011. Accessed: Dec. 09, 2024. [Online]. Available: <https://www.ncbi.nlm.nih.gov/books/NBK57145/>
- [22] H. Gao, J. Xiao, Y. Wei, H. Wang, H. Wan, and S. Liu, 'Regulation of Myogenic Differentiation by Topologically Microgrooved Surfaces for Skeletal Muscle Tissue Engineering', *ACS Omega*, vol. 6, no. 32, pp. 20931–20940, Aug. 2021, doi: 10.1021/acsomega.1c02347.
- [23] X. Du, L. Yang, and N. Liu, 'Recent Progress on Poly(3,4-Ethylenedioxythiophene):Poly(Styrenesulfonate) Bioelectrodes', *Small Sci.*, vol. 3, no. 7, p. 2300008, 2023, doi: 10.1002/smssc.202300008.
- [24] P.-A. Mouthuy *et al.*, 'Humanoid robots to mechanically stress human cells grown in soft bioreactors', *Commun. Eng.*, vol. 1, p. 2, May 2022, doi: 10.1038/s44172-022-00004-9.
- [25] F. M. Trovato, R. Imbesi, N. Conway, and P. Castrogiovanni, 'Morphological and Functional Aspects of Human Skeletal Muscle', *J. Funct. Morphol. Kinesiol.*, vol. 1, no. 3, Art. no. 3, Sep. 2016, doi: 10.3390/jfmk1030289.
- [26] A. Weisrock, R. Wüst, M. Olenic, P. Lecomte-Grosbras, and L. Thorrez, 'MyoFlnDer: An AI-Based Tool for Myotube Fusion Index Determination', *Tissue Eng. Part A*, vol. 30, no. 19–20, pp. 652–661, Oct. 2024, doi: 10.1089/ten.tea.2024.0049.



- [27] Y. Yoshimoto, M. Ikemoto-Uezumi, K. Hitachi, S. Fukada, and A. Uezumi, 'Methods for Accurate Assessment of Myofiber Maturity During Skeletal Muscle Regeneration', *Front. Cell Dev. Biol.*, vol. 8, Apr. 2020, doi: 10.3389/fcell.2020.00267.
- [28] J. A. Kim, Y. H. Shon, J. O. Lim, J. J. Yoo, H.-I. Shin, and E. K. Park, 'MYOD mediates skeletal myogenic differentiation of human amniotic fluid stem cells and regeneration of muscle injury', *Stem Cell Res. Ther.*, vol. 4, no. 6, p. 147, Dec. 2013, doi: 10.1186/scrt358.
- [29] M. Watanabe *et al.*, 'Blebbistatin, a myosin II inhibitor, suppresses contraction and disrupts contractile filaments organization of skinned taenia cecum from guinea pig', *Am. J. Physiol.-Cell Physiol.*, vol. 298, no. 5, pp. C1118–C1126, May 2010, doi: 10.1152/ajpcell.00269.2009.
- [30] H. Sakai *et al.*, 'Mechanisms of cisplatin-induced muscle atrophy', *Toxicol. Appl. Pharmacol.*, vol. 278, no. 2, pp. 190–199, Jul. 2014, doi: 10.1016/j.taap.2014.05.001.

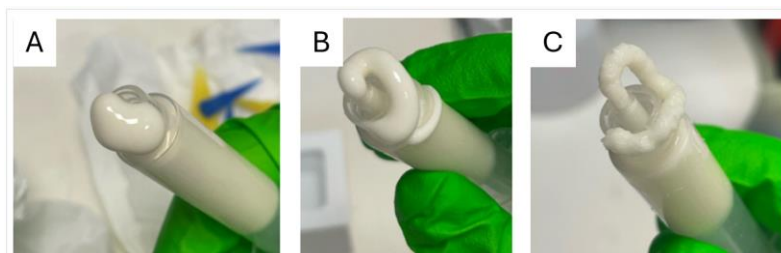
## 6. Appendix



**Fig. S1: Curing test**

**A.** PVA concentration (%) over heating time (min.), **B.** Curing tests performed following 0.5h, 1.0h, 1.5h, and 2.0h heating showing zoom-in images of (semi-)cured samples.

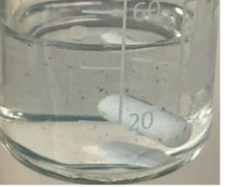
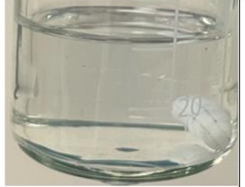
Following heating, the solution properties changed, with an initial smooth texture and subsequent drying and accumulation of air bubbles (Fig. S2A,B,C). Within 10 minutes following production, the mixture began to harden. By 40 minutes, extrusion became difficult due to encapsulated air. After one hour, extrusion was no longer possible. These observations suggested that a ~15-minute window post-heating should be used for optimal wet-spinning to avoid significant texture changes.



**Fig. S2: Post-heating PVA solution**

**A.** Immediately after heating showing smooth texture, **B.** 10 min. following heating showing change in texture, **C.** 40 min. following production showing drying of solution and accumulation of air bubbles.

### Pre- and post-heating dissolvability

|                                | 14kDa Mw PVA                                                                      | 18kDa Mw PVA                                                                       | 41kDa Mw PVA                                                                        |
|--------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Pre-heating<br>dissolvability  |  |  |  |
| Post-heating<br>dissolvability |  |  |  |

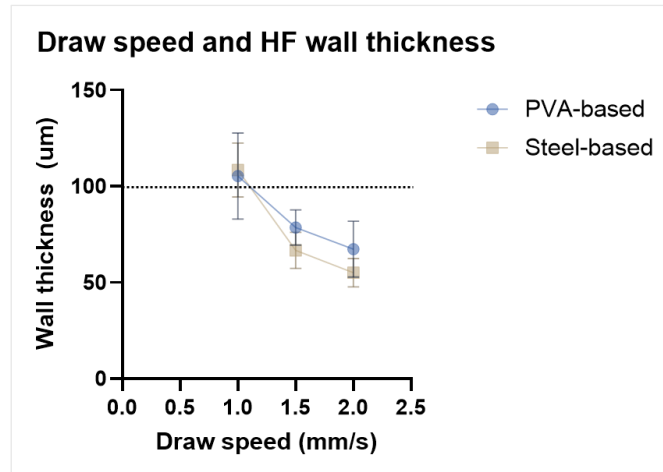
**Fig. S3: Dissolvability test**

Pre- and post-heating dissolvability of the fibre collector made from different PVA types. Pictures show collector dissolution following 6 min. in water under constant stirring.



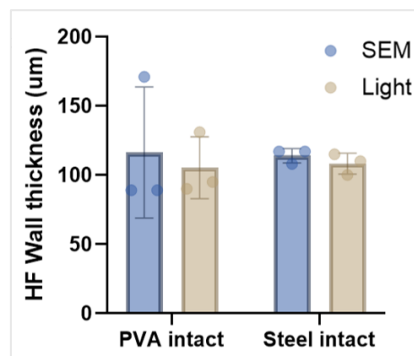
**Fig. S4: Instalments for minimised tension of the dissolvable fibre collector**

A tension controller (left) and a plastic tube (right) on the rotating spool of the electrospinner for reduced friction.



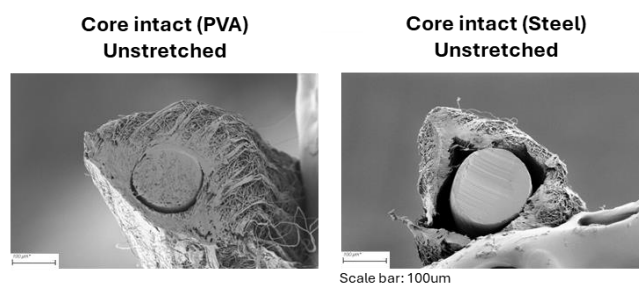
**Fig. S5: Draw speed and hollow fibre wall thickness**

Electrospinning draw speed and wall thickness of PVA- and steel-based electrospun hollow fibres (eHFs).



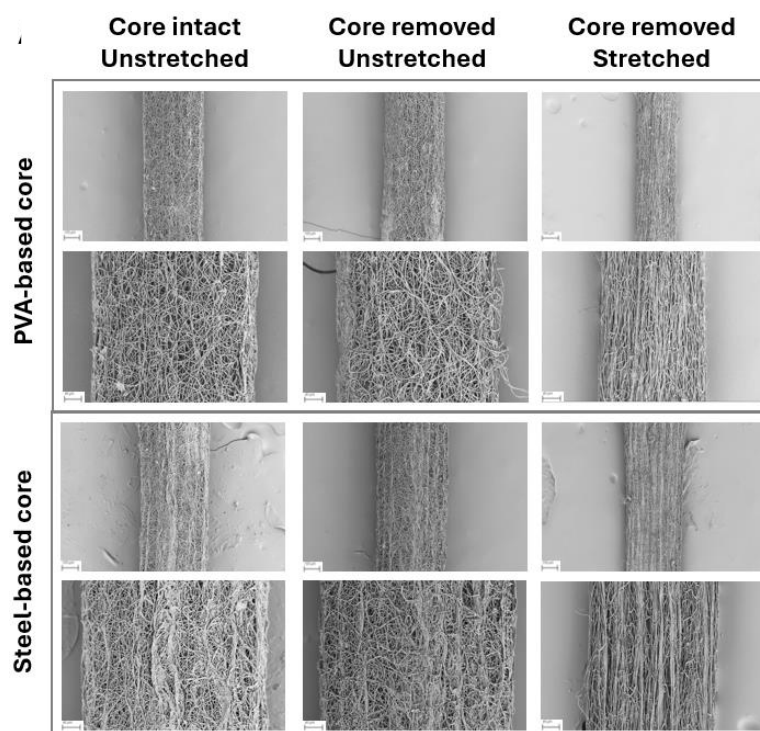
**Fig. S6: SEM vs light microscopy data for hollow fibre wall thickness**

Comparison of scanning electron microscopy (SEM) and light microscopy data of the wall thickness of PVA- and steel-based electrospun hollow fibres (eHFs).



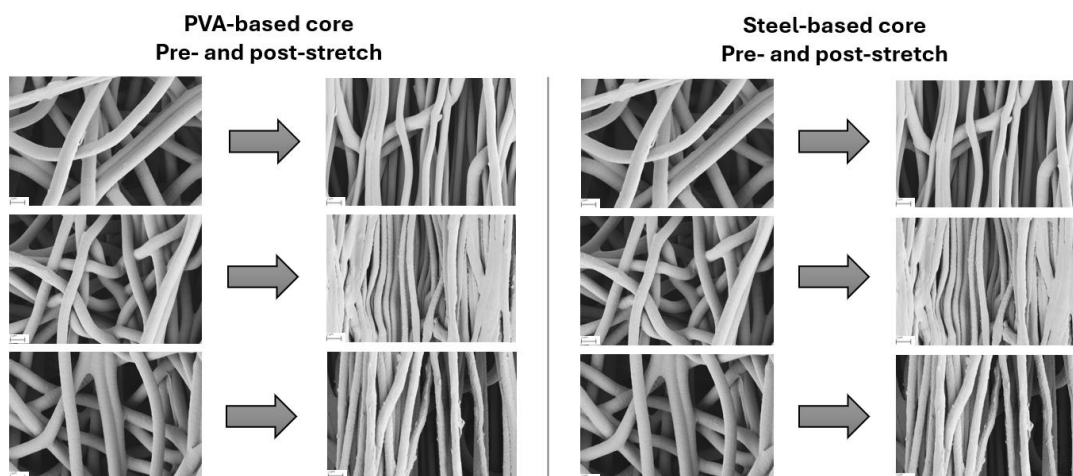
**Fig. S7: Hollow fibre cross sections**

Cross sections of unstretched electrospun hollow fibres (eHFs) with intact PVA and steel core, Scale bars: 100μm.



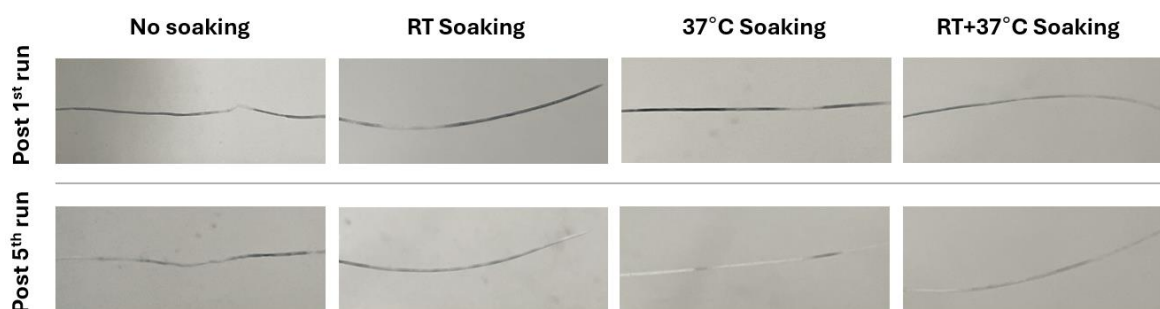
**Fig. S8: Electrospun hollow fibres**

Scanning electron microscopy (SEM) images of PVA- and steel-based electrospun hollow fibres (eHFs) with intact core unstretched, removed core unstretched, and removed core stretched, Scale bars: 40 $\mu$ m.



**Fig. S9: Stretching effects on electrospun microfibres**

Scanning electron microscopy (SEM) images of microfibres of pre- and post-stretch PVA- and steel-based electrospun hollow fibres (eHFs) at a magnification of 4.33k, Scale bars: 2 $\mu$ m.



**Fig. S10: Removal of collector core**

PVA-based electrospun hollow fibres (eHFs) following one and five 5-min. sonication runs at 40%amp with no pre-soaking, pre-soaking at room temperature (RT), pre-soaking at 37°C, and two sets of pre-soaking (RT, 37°C) and sonication. eHF material in white. Dissolvable collector material in black.